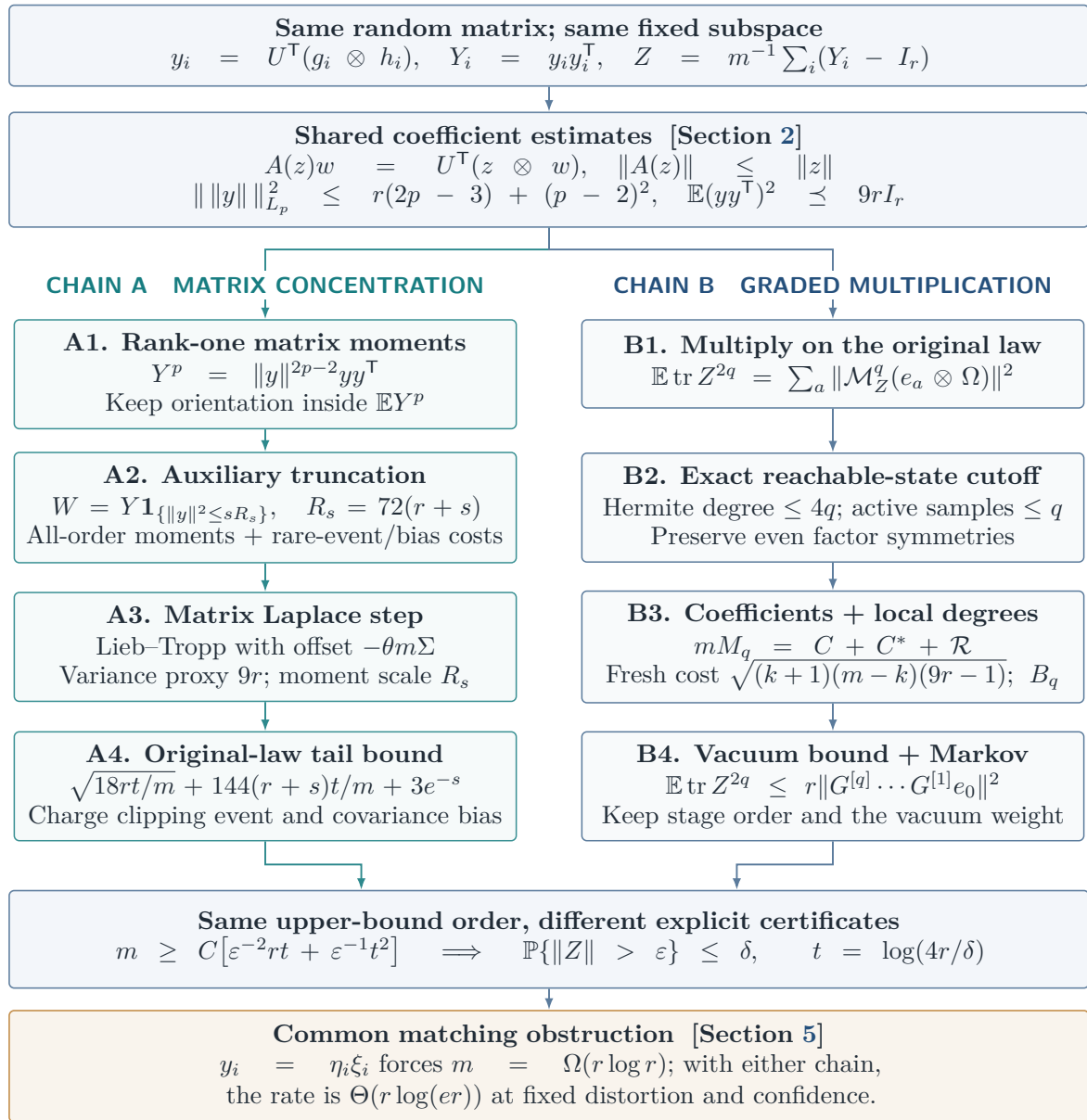


Sharp Gaussian Khatri–Rao embeddings

Two self-contained proof chains, one sharp fixed-confidence rate

The arguments share the same Gaussian sketch and collective coefficient estimates. Chain A then uses truncation and matrix concentration; Chain B uses exact graded multiplication and vacuum propagation. Both reach the same sample-size order. In the map below, full-width boxes are shared by both chains; the amber box is the common lower bound.



All application-specific steps and the matching lower bound are included here. Standard analytic inputs are stated explicitly and attributed. Numerical entries are sufficient theorem certificates, not empirical sample sizes. This research derivation has not been independently externally audited or formally verified.

1 Model, theorem, and comparison at a glance

1.1 The unchanged sketch and its quantifiers

Let $n_1, n_2, r, m \geq 1$ be integers with $r \leq n_1 n_2$, and fix a deterministic isometry

$$U : \mathbb{R}^r \longrightarrow \mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2}, \quad U^\top U = I_r.$$

For $1 \leq i \leq m$, let $g_i \sim N(0, I_{n_1})$ and $h_i \sim N(0, I_{n_2})$; all samples and both factors are mutually independent. Define

$$S = \frac{1}{\sqrt{m}} \begin{pmatrix} (g_1 \otimes h_1)^\top \\ \vdots \\ (g_m \otimes h_m)^\top \end{pmatrix}, \quad y_i = U^\top (g_i \otimes h_i), \quad (1.1)$$

$$Y_i = y_i y_i^\top, \quad X_i = Y_i - I_r, \quad Z = U^\top S^\top S U - I_r = \frac{1}{m} \sum_{i=1}^m X_i.$$

Independence gives $\mathbb{E}(g_i \otimes h_i)(g_i \otimes h_i)^\top = I_{n_1 n_2}$, so $\mathbb{E}Y_i = I_r$ and $\mathbb{E}X_i = 0$. The event $\|Z\| \leq \varepsilon$ is exactly the squared-norm embedding guarantee

$$(1 - \varepsilon)\|x\|^2 \leq \|S U x\|^2 \leq (1 + \varepsilon)\|x\|^2 \quad \text{for every } x \in \mathbb{R}^r.$$

Every probability statement below is for *each fixed* U , with constants uniform in U, n_1, n_2 . It is not a simultaneous statement for all subspaces after S is drawn. All logarithms are natural; unmarked matrix norms are operator norms.

Theorem 1.1 (Common endpoint). *There is a universal $C > 0$ such that, for $0 < \varepsilon, \delta < 1$ and $t = \log(4r/\delta)$,*

$$m \geq C \left(\frac{rt}{\varepsilon^2} + \frac{t^2}{\varepsilon} \right) \implies \mathbb{P}\{\|Z\| > \varepsilon\} \leq \delta. \quad (1.2)$$

Both proof chains below establish this upper bound independently after their shared local estimates. Conversely, for $U e_a = e_a \otimes e_1$ with $n_1 = r, n_2 = 2$, there is $c > 0$ such that uniformly over integers $1 \leq m \leq cr \log r$,

$$\mathbb{P}\{\|Z\| > 1/2\} \longrightarrow 1 \quad (r \longrightarrow \infty).$$

Therefore the worst-case sample order at distortion $1/2$ and success probability 0.99 is $\Theta(r \log(er))$.

The statement does not assert optimal joint dependence on ε and δ . The upper bound is proved for arbitrary subspace coefficients; only the lower-bound example uses a common factor.

1.2 Where the chains differ

Feature	Chain A: matrix concentration	Chain B: graded multiplication
Shared starting point	The exact map $A(z)w = U^T(z \otimes w)$, projected-row moments, and directional fourth moments.	
Object retained	One-row matrix powers $\mathbb{E}Y^p$.	The full original-law vacuum walk and its sample/degree sectors.
Cutoff	Auxiliary rare-outcome truncation $\ y\ ^2 \leq sR_s$; probability and bias are paid.	Exact Hermite reachability and active-sample cutoff; no outcome is removed.
Main assembly tool	Lieb’s trace concavity, iterated as the matrix Laplace method.	Fresh-sample and repeated-visit contractions, followed by nonnegative vacuum propagation.
Explicit output	Tail allowance in (3.12).	Untruncated moments; stationary, stage-dependent, and closed-form certificates.
Constants	One specified conservative baseline.	Closed form can be worse; stage-dependent certificates improve this baseline in the reported cases.
Additional consequence	Not claimed from its truncated certificate.	Exact $2q$ -wise sample-block moment transfer.

1.3 How to read this note

Section 2 proves the shared coefficient estimates. Sections 3 and 4 are the two upper-bound chains. Section 5 supplies the common lower bound; Section 6 compares fully specified constants and numerical certificates. Appendices give the Gaussian normalization, the constant-72 verification, and the numerical recurrence. No earlier project manuscript is needed to check an application-specific step.

The Gaussian logarithmic Sobolev and hypercontractive inequalities are attributed to the classical theory of Gross; short derivations are included in Appendix A. Chain A also uses Lieb’s trace-concavity theorem, stated explicitly in Section 3.3; that classical theorem is the one analytic input not reproved here. The matrix concentration step is derived from it, rather than quoted as a black box. Chain B does not use it.

For context, the order-two Gaussian specialization of Beretta–Musco’s Theorem 1.3 gives $O(\varepsilon^{-2}r \log^3(r/(\varepsilon\delta)))$ rows [6]. Their theorem has a broader distributional and tensor-order scope. This note compares only the real Gaussian order-two case and makes no publication-priority claim.

2 Shared coefficient estimates

The same subspace geometry feeds both chains. Keep the map $A(z)$ collected before taking norms: its operator-norm Lipschitz constant is 1, not $\sqrt{r}$. Then keep the rank-one orientation when a matrix-valued estimate is needed.

2.1 Gaussian moment control with explicit normalization

Write γ_N for standard Gaussian measure on $\mathbb{R}^N$ and $\text{Ent}(F) = \mathbb{E}[F \log F] - (\mathbb{E}F) \log \mathbb{E}F$ for $F \geq 0$. The Gaussian logarithmic Sobolev inequality is

$$\text{Ent}_{\gamma_N}(f^2) \leq 2\mathbb{E}_{\gamma_N} \|\nabla f\|^2. \quad (2.1)$$

For a Hilbert-valued homogeneous Hermite polynomial F_d of total degree d , Gaussian hypercontractivity gives

$$\|F_d\|_{L_u} \leq (u-1)^{d/2} \|F_d\|_{L_2}, \quad u \geq 2. \quad (2.2)$$

The normalization and short semigroup proofs appear in Appendix A; these are classical inequalities [2, 3].

Lemma 2.1 (Lipschitz moments). *For a nonnegative L -Lipschitz function f on a finite-dimensional standard Gaussian space and any real $p \geq 2$,*

$$\|f\|_{L_p}^2 \leq \|f\|_{L_2}^2 + (p-2)L^2. \quad (2.3)$$

Proof. Assume first that f is smooth, bounded, and bounded below by a positive constant, and set $M(p) = \mathbb{E}f^p$.

1. Differentiate the running moment in its exponent.

Since $\frac{d}{dp} \mathbb{E}f^p = \mathbb{E}f^p \log f$, a direct computation gives

$$\frac{d}{dp} M(p)^{2/p} = \frac{2}{p^2} M(p)^{2/p-1} \text{Ent}(f^p).$$

2. Bound the entropy by the logarithmic Sobolev inequality.

Apply (2.1) to $f^{p/2}$ and use $\|\nabla f\| \leq L$:

$$\text{Ent}(f^p) \leq \frac{p^2 L^2}{2} \mathbb{E}f^{p-2}.$$

Hölder's inequality gives $\mathbb{E}f^{p-2} \leq M(p)^{(p-2)/p}$, so the two previous displays combine into

$$\frac{d}{dp} M(p)^{2/p} \leq L^2 M(p)^{2/p-1} M(p)^{1-2/p} = L^2.$$

3. Integrate from 2 to p .

This yields $M(p)^{2/p} \leq M(2) + (p-2)L^2$, which is (2.3). Smoothing, a positive regularization, and truncation of the scalar test function justify the same bound for Lipschitz f by passage to the limit. $\square$

2.2 The collective conditional Gaussian map

Reshape the columns of U as matrices $B_1, \dots, B_r \in \mathbb{R}^{n_1 \times n_2}$. Then

$$\text{tr}(B_a^\top B_b) = \delta_{ab}, \quad y_a = g^\top B_a h. \quad (2.4)$$

For $z \in \mathbb{R}^{n_1}$ define $A(z) : \mathbb{R}^{n_2} \rightarrow \mathbb{R}^r$ by

$$A(z)w = U^\top(z \otimes w), \quad y = A(g)h. \quad (2.5)$$

Because $U^\top$ is a contraction,

$$\|A(z)\| \leq \|z\|, \quad \mathbb{E}\|A(g)\|_F^2 = r, \quad \|A(z)\|_F^2 \leq r\|z\|^2. \quad (2.6)$$

The second assertion follows by summing $\|B_a\|_F^2 = 1$; the third follows from $\sum_a \|B_a^\top z\|^2 \leq \sum_a \|B_a\|_F^2 \|z\|^2$. Linearity of A makes $z \mapsto \|A(z)\|$ 1-Lipschitz and $z \mapsto \|A(z)\|_F$ $\sqrt{r}$ -Lipschitz.

Lemma 2.2 (Projected-row moments). *For every real $p \geq 2$,*

$$\boxed{\| \|y\| \|_{L_p}^2 \leq r(2p-3) + (p-2)^2.} \quad (2.7)$$

In particular $\| \|y\| \|_{L_p} \leq \sqrt{2rp} + p$.

Proof. 1. Bound the moments of the two Lipschitz norms of $A(g)$.

By (2.6), $z \mapsto \|A(z)\|_F$ is $\sqrt{r}$ -Lipschitz with squared L_2 norm r , and $z \mapsto \|A(z)\|$ is 1-Lipschitz with squared L_2 norm at most r . Lemma 2.1 gives

$$\| \|A(g)\|_F \|_{L_p}^2 \leq r(p-1), \quad \| \|A(g)\| \|_{L_p}^2 \leq r + p - 2.$$

2. Condition on g and integrate over h .

For fixed g , the function $h \mapsto \|A(g)h\|$ has Lipschitz constant $\|A(g)\|$ and squared L_2 norm $\|A(g)\|_F^2$. Lemma 2.1, applied in the variable h alone, gives

$$(\mathbb{E}_h \|A(g)h\|^p)^{2/p} \leq \|A(g)\|_F^2 + (p-2)\|A(g)\|^2.$$

3. Combine the two factor bounds.

Take $L_{p/2}$ norms in g on both sides. Minkowski's inequality in $L_{p/2}$ separates the two summands on the right, and Step 1 bounds each of them:

$$\| \|y\| \|_{L_p}^2 \leq r(p-1) + (p-2)(r + p - 2) = r(2p-3) + (p-2)^2.$$

All coefficients remain collected. No common Schmidt basis and no independence of the coordinates of y is assumed. $\square$

2.3 Directional fourth moments and matrix variance

Lemma 2.3 (Fourth moments). *For every $x \in \mathbb{R}^r$,*

$$\mathbb{E}\langle x, y \rangle^4 \leq 9\|x\|^4. \quad (2.8)$$

Consequently, with $Y = yy^\top$ and $X = Y - I_r$,

$$\mathbb{E}Y^2 \preceq 9rI_r, \quad \mathbb{E}X^2 \preceq v_r I_r, \quad v_r := 9r - 1. \quad (2.9)$$

Proof. 1. The directional fourth moment.

For unit x , put $B_x = \sum_a x_a B_a$. Equation (2.4) gives $\|B_x\|_F = 1$. Conditional on g , $g^\top B_x h$ is Gaussian with variance $g^\top Q g$, where $Q = B_x B_x^\top \succeq 0$ and $\text{tr } Q = 1$. Diagonalizing Q gives

$$\mathbb{E}\langle x, y \rangle^4 = 3\mathbb{E}(g^\top Q g)^2 = 3(\text{tr } Q)^2 + 6 \text{tr } Q^2 \leq 9.$$

Homogeneity proves (2.8).

2. The matrix second moment.

For unit x , Cauchy–Schwarz in each summand gives

$$x^\top (\mathbb{E} Y^2) x = \sum_{a=1}^r \mathbb{E}[y_a^2 \langle x, y \rangle^2] \leq \sum_{a=1}^r (\mathbb{E} y_a^4)^{1/2} (\mathbb{E} \langle x, y \rangle^4)^{1/2} \leq 9r.$$

3. Center.

Finally $\mathbb{E} X^2 = \mathbb{E} Y^2 - 2\mathbb{E} Y + I_r = \mathbb{E} Y^2 - I_r$. □

The common-factor case $y = \eta \xi$ has $\|y\|_{L_p} = \|\eta\|_{L_p} \|\xi\|_{L_p}$, of order $\sqrt{rp} + p$. It calibrates the uniform order of Lemma 2.2, but is not used to restrict its upper-bound hypotheses.

3 Chain A: matrix moments, truncation, and concentration

Chain A never constructs a graded multiplication operator. It converts the common row estimates into operator-valued moments of one rank-one matrix, truncates only for the exponential series, and uses matrix Laplace concentration. The final statement is for the original, unmodified sketch.

3.1 A1. Keep matrix orientation when estimating powers

Lemma 3.1 (Explicit one-row matrix moments). *For $Y = yy^\top$ and every integer $p \geq 3$,*

$$\mathbb{E}Y^p \preceq 3[(8p - 11)r + (4p - 6)^2]^{p-1}I_r, \quad (3.1)$$

$$\mathbb{E}Y^p \preceq \frac{p!}{2} 9r [72(r + p)]^{p-2}I_r. \quad (3.2)$$

For $p = 2$, $\mathbb{E}Y^2 \preceq 9rI_r$.

Proof. **1. Bound each directional moment of the matrix power.**

The rank-one identity

$$Y^p = \|y\|^{2p-2}yy^\top$$

lets us keep the direction x until after expectation. For any unit x ,

$$\begin{aligned} x^\top(\mathbb{E}Y^p)x &= \mathbb{E}[\|y\|^{2p-2}\langle x, y \rangle^2] \\ &\leq (\mathbb{E}\|y\|^{4p-4})^{1/2}(\mathbb{E}\langle x, y \rangle^4)^{1/2} \\ &\leq 3\|\|y\|\|_{L_{4p-4}}^{2p-2} \\ &\leq 3[(8p - 11)r + (4p - 6)^2]^{p-1}. \end{aligned}$$

The second line is Cauchy–Schwarz; the third bounds the directional factor by Lemma 2.3; the last applies Lemma 2.2 at exponent $4p - 4$. Since x is arbitrary, this proves (3.1) as a Loewner bound.

2. Pass to the uniform moment scale $72(r + p)$.

The scalar inequality

$$3[(8p - 11)r + (4p - 6)^2]^{p-1} \leq \frac{p!}{2} 9r [72(r + p)]^{p-2} \quad (3.3)$$

is proved for every $r \geq 1, p \geq 3$ in Appendix B, giving (3.2). The $p = 2$ case is Lemma 2.3. $\square$

The distinction is between $\|\mathbb{E}Y^p\|$ and $\mathbb{E}\|Y\|^p$. Replacing Y by its norm before taking expectation discards the direction used above.

3.2 A2. Truncate without paying the full cutoff height

The untruncated Y need not have a finite positive exponential moment. For example, when $r = 1$ and $y = gh$, $Y = g^2h^2$ has infinite moment-generating function at every positive argument. Conditional on g , the Gaussian integral in h already diverges whenever $2\theta g^2 \geq 1$, an event of positive probability. Thus a positive-matrix exponential theorem cannot be applied to Y directly.

Lemma 3.2 (All-order moments of a bounded auxiliary matrix). *Fix an integer $s \geq 2$ and define*

$$R_s = 72(r + s), \quad L_s = sR_s, \quad W = Y\mathbf{1}_{\{\|y\|^2 \leq L_s\}}, \quad \Sigma = \mathbb{E}W. \quad (3.4)$$

Then

$$\mathbb{P}\{W \neq Y\} \leq e^{-2s}, \quad (3.5)$$

$$0 \preceq I_r - \Sigma \preceq 3e^{-s} I_r, \quad (3.6)$$

$$\mathbb{E}W^p \preceq \frac{p!}{2} 9r R_s^{p-2} I_r \quad \text{for every integer } p \geq 2. \quad (3.7)$$

Proof. **1. Bound the event on which the sample is changed.**

At exponent $2s$, Lemma 2.2 gives

$$\| \|y\|^2 \|_{L_s} \leq r(4s - 3) + (2s - 2)^2 \leq 4s(r + s).$$

Markov's inequality therefore gives

$$\mathbb{P}\{\|y\|^2 > L_s\} \leq \left(\frac{4s(r + s)}{72s(r + s)} \right)^s = 18^{-s} \leq e^{-2s}.$$

2. Charge the covariance bias in each direction.

For a unit vector x , Lemma 2.3 and Cauchy–Schwarz imply

$$x^\top (I_r - \Sigma)x = \mathbb{E}[\langle x, y \rangle^2 \mathbf{1}_{\{\|y\|^2 > L_s\}}] \leq 3\mathbb{P}\{\|y\|^2 > L_s\}^{1/2} \leq 3e^{-s}.$$

The integrand is nonnegative, proving both sides of (3.6).

3. Prove the matrix moment bound through order s .

For $p = 2$, $\mathbb{E}W^2 \preceq \mathbb{E}Y^2 \preceq 9r I_r$. For $3 \leq p \leq s$, the scalar event in (3.4) gives $W^p \preceq Y^p$ pointwise. Lemma 3.1 and $p \leq s$ yield

$$\mathbb{E}W^p \preceq \frac{p!}{2} 9r [72(r + p)]^{p-2} I_r \preceq \frac{p!}{2} 9r R_s^{p-2} I_r.$$

4. Extend to every order required by the exponential series.

For an integer $p > s$, spectral calculus on $0 \preceq W \preceq L_s I_r$ gives

$$W^p \preceq L_s^{p-s} W^s.$$

Consequently,

$$\mathbb{E}W^p \preceq \frac{s!}{2} 9r R_s^{s-2} L_s^{p-s} I_r = \frac{s! s^{p-s}}{2} 9r R_s^{p-2} I_r \preceq \frac{p!}{2} 9r R_s^{p-2} I_r.$$

The last inequality follows by comparing each factor of $(s + 1)(s + 2) \cdots p$ with s . This proves all of (3.7), not just the first s moments. $\square$

The almost-sure cutoff height is $L_s = sR_s$, but the matrix moment-growth scale is only R_s . Charging L_s as the bounded-summand scale would lose the factor s .

3.3 A3. A matrix Laplace step with valid centering

The classical matrix-analysis input is Lieb's trace-concavity theorem: for a fixed self-adjoint H , the map

$$Q \mapsto \text{tr} \exp(H + \log Q) \quad (3.8)$$

is concave on the positive-definite cone [4, 5]. We use no assumption that a random matrix commutes with its expectation.

Lemma 3.3 (Matrix Laplace iteration with an offset). *For independent self-adjoint random matrices $A_1, \dots, A_m$ with finite matrix exponential moments, and deterministic self-adjoint H ,*

$$\mathbb{E} \operatorname{tr} \exp \left(H + \sum_{i=1}^m A_i \right) \leq \operatorname{tr} \exp \left(H + \sum_{i=1}^m \log \mathbb{E} e^{A_i} \right). \quad (3.9)$$

Proof. For one random matrix A , write $Q = e^A$. Jensen’s inequality applied to (3.8) gives

$$\mathbb{E} \operatorname{tr} \exp(H + A) \leq \operatorname{tr} \exp(H + \log \mathbb{E} e^A).$$

For the sum, condition first on $A_1, \dots, A_{m-1}$, treating all their terms as part of H . Independence makes $\log \mathbb{E} e^{A_m}$ deterministic. Repeat for $A_{m-1}, \dots, A_1$. This is the matrix Laplace iteration used by Tropp; no reordering of products or commuting-mean identity is involved [5]. $\square$

Lemma 3.4 (Concentration from positive matrix moments). *Let $W_1, \dots, W_m$ be independent, identically distributed bounded positive semidefinite $r \times r$ matrices with $\mathbb{E} W_i = \Sigma$. Suppose that for $p \geq 2$,*

$$\mathbb{E} W_i^p \preceq \frac{p!}{2} v R^{p-2} I_r.$$

Then for every $u > 0$,

$$\mathbb{P} \left\{ \left\| \frac{1}{m} \sum_i W_i - \Sigma \right\| > \sqrt{\frac{2vu}{m}} + \frac{2Ru}{m} \right\} \leq 2re^{-u}. \quad (3.10)$$

Proof. 1. Bound the logarithm of the uncentered exponential moment.

For $|\theta|R < 1$, positivity of each W_i^p and the exponential series imply

$$\mathbb{E} e^{\theta W_i} - I_r - \theta \Sigma \preceq \sum_{p \geq 2} \frac{|\theta|^p}{p!} \mathbb{E} W_i^p \preceq \frac{\theta^2 v}{2(1 - |\theta|R)} I_r.$$

For positive definite Q , spectral calculus gives $\log Q \preceq Q - I_r$. Therefore

$$\log \mathbb{E} e^{\theta W_i} \preceq \theta \Sigma + \frac{\theta^2 v}{2(1 - |\theta|R)} I_r. \quad (3.11)$$

2. Center by a deterministic offset inside the trace exponential.

In Lemma 3.3, take $A_i = \theta W_i$ and $H = -\theta m \Sigma$. Using (3.11),

$$\mathbb{E} \operatorname{tr} \exp \left(\theta \sum_i (W_i - \Sigma) \right) \leq r \exp \left(\frac{m \theta^2 v}{2(1 - |\theta|R)} \right).$$

Here a Loewner bound on the exponent gives a bound on each eigenvalue and hence on the trace exponential. No operator monotonicity of the exponential is needed.

3. Convert the exponential bound to the two spectral tails.

For a self-adjoint D and $\theta > 0$, the event $\lambda_{\max}(D) > x$ implies $\operatorname{tr} e^{\theta D} > e^{\theta x}$. Markov, followed by $\theta = x/(mv + Rx)$, gives

$$\mathbb{P} \left\{ \lambda_{\max} \left(\sum_i (W_i - \Sigma) \right) > x \right\} \leq r \exp \left(-\frac{x^2}{2(mv + Rx)} \right).$$

Use negative θ for the lower spectral tail and add the probabilities. Finally $x = \sqrt{2mvu} + 2Ru$ satisfies $x^2 \geq 2u(mv + Rx)$, proving (3.10). $\square$

3.4 A4. Restore the original samples and solve for m

Theorem 3.5 (Explicit Chain A certificate). *Let $t = \log(4r/\delta)$ and $s = \lceil \log(4m/\delta) \rceil$. Then*

$$\mathbb{P} \left\{ \|Z\| > \sqrt{\frac{18rt}{m}} + \frac{144(r+s)t}{m} + 3e^{-s} \right\} \leq \delta. \quad (3.12)$$

In particular, having the displayed allowance at most ε is an explicit sufficient condition for the embedding guarantee.

Proof. The integer s is at least 2.

1. Replace every sample by its truncation.

Apply Lemma 3.2 separately to the independent samples, with the same deterministic cutoff. A union bound gives

$$\mathbb{P}\{\exists i : W_i \neq Y_i\} \leq me^{-2s} \leq \frac{\delta^2}{16m} \leq \frac{\delta}{16}.$$

2. Concentrate the truncated average.

Lemma 3.4, with $v = 9r$, $R = R_s = 72(r+s)$, and $u = t$, fails with probability at most $2re^{-t} = \delta/2$.

3. Return to the original sketch.

On the intersection of the two good events, every W_i equals Y_i , so

$$\begin{aligned} \|Z\| &\leq \left\| \frac{1}{m} \sum_i W_i - \Sigma \right\| + \|\Sigma - I_r\| \\ &\leq \sqrt{\frac{18rt}{m}} + \frac{144(r+s)t}{m} + 3e^{-s}. \end{aligned}$$

The total failure probability is less than δ . The conclusion concerns the original S , not an estimator that clips its rows during execution. $\square$

Proposition 3.6 (The common sample order follows from Chain A). *The explicit certificate (3.12) implies (1.2) with a universal constant.*

Proof. **1. Simplify the allowance to two monotone terms.**

Since $s \leq \log(4m/\delta) + 1$ and $3e^{-s} \leq 3\delta/(4m)$, the allowance in (3.12) is at most

$$C_1 \left[\sqrt{\frac{rt}{m}} + \frac{(r + \log(4m/\delta) + 1)t}{m} \right] \quad (3.13)$$

for an absolute C_1 . Both terms decrease with real $m \geq 1$; for the second, differentiate $(r + \log(4m/\delta) + 1)/m$.

2. Evaluate at the candidate sample size.

Set

$$m_0 = K \left(\frac{rt}{\varepsilon^2} + \frac{t^2}{\varepsilon} \right) = K\varepsilon^{-2}t(r + \varepsilon t), \quad K \geq 1.$$

Here $t > 1$, $r \leq e^t$, and $t \leq e^t$, so

$$\log(4m_0/\delta) + 1 \leq \log K + 2\log(1/\varepsilon) + 3t + \log 2 + 1.$$

3. Bound each term and fix the constant.

First,

$$\sqrt{\frac{rt}{m_0}} \leq \frac{\varepsilon}{\sqrt{K}}.$$

Second,

$$\begin{aligned} \frac{(r + \log(4m_0/\delta) + 1)t}{m_0} &\leq \frac{\varepsilon^2}{K} \frac{r + 3t + 2\log(1/\varepsilon) + \log K + \log 2 + 1}{r + \varepsilon t} \\ &\leq \frac{C_2(1 + \log K)}{K} \varepsilon. \end{aligned}$$

For the last step, use $r \geq 1$, $r + \varepsilon t \geq \varepsilon t$, and $\varepsilon \log(1/\varepsilon) \leq 1/e$. Choose K so that $C_1[K^{-1/2} + C_2(1 + \log K)/K] \leq 1$. Monotonicity of (3.13) handles every larger integer m . $\square$

4 Chain B: exact graded multiplication and vacuum propagation

Chain B uses the four core stages of the framework: exact multiplication, reachable graded states, coefficient-sensitive contractions, and vacuum-moment propagation. No row is clipped and no matrix Laplace theorem is invoked. The central estimate charges each already-active sample its own local degree *before* summing over samples.

4.1 B1. The original-law multiplication identity

Let

$$\mathcal{H} = \mathbb{R}^r \otimes \bigotimes_{i=1}^m L_2(\gamma_{n_1} \otimes \gamma_{n_2}), \quad \Omega = 1.$$

The external factor $\mathbb{R}^r$ carries the matrix indices, one L_2 factor carries the randomness of one sample, and the vacuum Ω is the constant function 1 in every sample register. Define $\mathcal{M}_Z f = Zf$ on the vector polynomials in this Hilbert space. For an external standard basis vector e_a ,

$$\mathcal{M}_Z^j(e_a \otimes \Omega)(g_1, h_1, \dots, g_m, h_m) = Z^j e_a.$$

All of these polynomials are square integrable. Because the entries of Z are real random variables, $\mathcal{M}_Z$ is self-adjoint on this domain, so half of the $2q$ factors can be moved to each side of the vacuum inner product:

$$\mathbb{E} e_a^\top Z^{2q} e_a = \langle e_a \otimes \Omega, \mathcal{M}_Z^{2q}(e_a \otimes \Omega) \rangle_{\mathcal{H}} = \|\mathcal{M}_Z^q(e_a \otimes \Omega)\|_{\mathcal{H}}^2.$$

Summing over a gives the identity the chain rests on:

$$\boxed{\mathbb{E} \operatorname{tr} Z^{2q} = \sum_{a=1}^r \|\mathcal{M}_Z^q(e_a \otimes \Omega)\|_{\mathcal{H}}^2.} \quad (4.1)$$

The factor r that will eventually arise counts external vacuum seeds. We do not take a trace over the generally much larger Hermite occupation space. Every visit to sample i still uses the same g_i, h_i .

4.2 B2. Hermite degree, active samples, and the exact cutoff

Let $h_\ell = \operatorname{He}_\ell / \sqrt{\ell!}$ denote the normalized probabilists' Hermite polynomial. Its recurrence is

$$x h_\ell(x) = \sqrt{\ell+1} h_{\ell+1}(x) + \sqrt{\ell} h_{\ell-1}(x), \quad h_{-1} = 0. \quad (4.2)$$

Products of these functions form an orthonormal basis in each Gaussian sample space. Let N_i record the total local degree in (g_i, h_i) . Let P_i be the orthogonal projection onto the local constant function, tensored with identities on all other factors, and put $Q_i = I - P_i$.

Definition 4.1 (An active sample). A sample is active in a product Hermite component when its local factor is nonconstant, equivalently when it lies in $Q_i L_2$. The exact active set is the set of such indices. Its size is a second grading, distinct from total degree.

An active set is not a history of labels that have ever occurred. A sample can return to its local vacuum, and then be activated again by a later multiplication.

There are exact sign-flip symmetries. Replacing the entire vector g_i by $-g_i$ changes y_i to $-y_i$ but leaves Y_i , X_i , and Z unchanged; the same holds for h_i . We restrict to functions having even total Hermite degree separately in every g_i and every h_i register. The vacuum belongs to this subspace, and multiplication by Z preserves it.

After j multiplications from vacuum,

$$\sum_i N_i \leq 4j, \quad \#\{\text{active samples}\} \leq j. \quad (4.3)$$

Indeed, X_i has polynomial degree at most four and acts on only one sample.

Let Π_q project onto the symmetry subspace with total degree at most $4q$ and at most q active samples, and set

$$M_q = \Pi_q \mathcal{M}_Z \Pi_q. \quad (4.4)$$

This finite-dimensional operator is self-adjoint. Inducting on j using (4.3) gives

$$M_q^j(e_a \otimes \Omega) = \mathcal{M}_Z^j(e_a \otimes \Omega), \quad 0 \leq j \leq q. \quad (4.5)$$

Thus (4.1) remains exact with M_q in place of $\mathcal{M}_Z$. This is the finite-moment compression principle of the framework, proved here for the present model [1]. No bound on the full unbounded Gaussian multiplication operator is required.

4.3 B3. Split activation, return, and repeated visits

Write $\mathcal{X}_i$ for multiplication by X_i . Centering is the exact identity $P_i \mathcal{X}_i P_i = 0$. Therefore

$$\mathcal{X}_i = Q_i \mathcal{X}_i P_i + P_i \mathcal{X}_i Q_i + Q_i \mathcal{X}_i Q_i.$$

Define, on the common cutoff,

$$C = \Pi_q \sum_i Q_i \mathcal{X}_i P_i \Pi_q, \quad \mathcal{R} = \Pi_q \sum_i Q_i \mathcal{X}_i Q_i \Pi_q. \quad (4.6)$$

Then

$$M_q = \frac{1}{m}(C + C^* + \mathcal{R}). \quad (4.7)$$

The operator C increases active-sample count by one; C^* decreases it by one; $\mathcal{R}$ preserves the exact active set. These are actual full-space compressions, so using adjointness is legitimate. Let $\mathcal{K}_k$ denote the cutoff sector with exactly k active samples.

4.3.1 Activation and return are controlled by matrix variance

Lemma 4.2 (Fresh-sample contraction). *With $v_r = 9r - 1$, for $0 \leq k < \min(q, m)$,*

$$\|C : \mathcal{K}_k \rightarrow \mathcal{K}_{k+1}\| \leq \sqrt{(k+1)(m-k)} v_r. \quad (4.8)$$

The reverse map C^ has the same bound.*

Proof. 1. Compute the single-insertion Gram before the output cutoff.

For a sample currently in its local vacuum,

$$\begin{aligned} (Q_i \mathcal{X}_i P_i)^*(Q_i \mathcal{X}_i P_i) &= P_i \mathcal{X}_i Q_i \mathcal{X}_i P_i \\ &= P_i \mathcal{X}_i^2 P_i = (\mathbb{E} X_i^2) P_i \preceq v_r P_i. \end{aligned}$$

The middle equality uses $P_i \mathcal{X}_i P_i = 0$. It remains valid with arbitrary Hilbert-valued coefficients in the other sample registers. Output projection by Π_q can only decrease a norm.

2. Count the inputs that can meet at one output set.

Decompose $f = \sum_{|S|=k} f_S$ by exact active set; these inputs are orthogonal. For an output set T of size $k+1$, the contributing inputs are $f_{T \setminus \{i\}}$, one for each $i \in T$. Cauchy–Schwarz gives

$$\left\| \sum_{i \in T} \Pi_q Q_i \mathcal{X}_i P_i f_{T \setminus \{i\}} \right\|^2 \leq (k+1) v_r \sum_{i \in T} \|f_{T \setminus \{i\}}\|^2.$$

Sum over T . Each input set S appears once for each of the $m-k$ absent labels, yielding

$$\|Cf\|^2 \leq (k+1)(m-k) v_r \sum_{|S|=k} \|f_S\|^2.$$

Take square roots and then use adjointness. $\square$

4.3.2 Repeated visits are controlled by local degree

For one sample define the row multiplier

$$Tf = y^\top f, \quad T^*T = \mathcal{M}_Y$$

as a quadratic-form identity on polynomials. All following estimates allow a Hilbert-valued coefficient space for the other registers. Hölder is applied only in the current sample variables, with that coefficient space kept intact.

Lemma 4.3 (Local degree-dependent row contraction). *Let f_d have homogeneous local Hermite degree $d \geq 1$. For every real $\lambda > 0$,*

$$\|Tf_d\|_2^2 \leq e^{2/\lambda} [(4\lambda d + 1)r + 4\lambda^2 d^2] \|f_d\|_2^2. \quad (4.9)$$

For a finite sum $f = \sum_{d \geq 1} f_d$ with no local vacuum component,

$$\|Tf\|_2^2 \leq 3e^{2/\lambda} \sum_{d \geq 1} [(4\lambda d + 1)r + 4\lambda^2 d^2] \|f_d\|_2^2. \quad (4.10)$$

Proof. 1. Choose the Hölder exponent from the actual local degree.

Set $p = \lambda d + 1 > 1$. Hölder and (2.2) give

$$\begin{aligned} \|y^\top f_d\|_2 &\leq \| \|y\| \|f_d\|_{L_{2p/(p-1)}} \|f_d\|_{L_{2p/(p-1)}} \\ &\leq \| \|y\| \|_{L_{2p}} \left(1 + \frac{2}{\lambda d}\right)^{d/2} \|f_d\|_2. \end{aligned}$$

Lemma 2.2 bounds the square of the first factor by

$$r(4p-3) + (2p-2)^2 = (4\lambda d + 1)r + 4\lambda^2 d^2.$$

Since $(1 + 2/(\lambda d))^d \leq e^{2/\lambda}$, this proves (4.9). The multiplier and the polynomial use the same sample; no independence between y and f_d is assumed.

2. Sum homogeneous inputs using their output-degree overlap.

Every coordinate of y is a pure second-chaos polynomial. In fact it is a linear combination of $g_a h_b$, with distinct factor coordinates. Write $\mathcal{F}_e$ for the closed span of the components of exact local Hermite degree e , with arbitrary coefficients in the other registers. Applying the Hermite recurrence (4.2) twice shows

$$Tf_d \in \mathcal{F}_{d-2} \oplus \mathcal{F}_d \oplus \mathcal{F}_{d+2}.$$

At any fixed output degree, at most three input degrees contribute. Cauchy–Schwarz on those contributions, followed by orthogonality of the output degrees, gives $\|Tf\|_2^2 \leq 3 \sum_d \|Tf_d\|_2^2$. Now use (4.9). $\square$

Define the deterministic allowance

$$B_q(r; \lambda) = 3e^{2/\lambda} [(16\lambda + 1)rq + 64\lambda^2 q^2]. \quad (4.11)$$

The positive real λ is a proof parameter, not a property of the subspace. In particular,

$$B_q(r; 2) = 99e r q + 768e q^2, \quad B_q(r; 1) = 51e^2 r q + 192e^2 q^2. \quad (4.12)$$

Lemma 4.4 (Sum of already-active sample interactions). *On the entire cutoff of (4.4),*

$$\|\mathcal{R}\| \leq B_q(r; \lambda). \quad (4.13)$$

Its restriction to the vacuum sector is zero.

Proof. 1. Fix the exact active set and expose the positive part.

On a support sector S with $k = |S| \leq q$, the quadratic form is

$$\langle f, \mathcal{R}f \rangle = \sum_{i \in S} \|T_i f\|_2^2 - k \|f\|_2^2. \quad (4.14)$$

This follows because $Q_i f = f$ for $i \in S$ and $Q_i f = 0$ otherwise, together with the one-sample identity $\langle f, \mathcal{X}_i f \rangle = \langle f, (\mathcal{M}_{Y_i} - I)f \rangle = \|T_i f\|_2^2 - \|f\|_2^2$. Thus $\mathcal{R} \succeq -kI$ on this sector.

2. Sum each label's actual degree before relaxing any budget.

Let $f = \sum_{\mathbf{d}} f_{\mathbf{d}}$ be the joint local-degree decomposition. The number operators N_i commute, and their joint eigenspaces are orthogonal. Apply (4.10) to each $i \in S$ and then combine the sums:

$$\begin{aligned} \sum_{i \in S} \|T_i f\|_2^2 &\leq 3e^{2/\lambda} \sum_{\mathbf{d}} \left[r \left(4\lambda \sum_{i \in S} d_i + k \right) + 4\lambda^2 \sum_{i \in S} d_i^2 \right] \|f_{\mathbf{d}}\|_2^2 \\ &\leq 3e^{2/\lambda} [(16\lambda + 1)rq + 64\lambda^2 q^2] \|f\|_2^2. \end{aligned}$$

The last inequality uses precisely

$$\sum_i d_i \leq 4q, \quad k \leq q, \quad \sum_i d_i^2 \leq \left(\sum_i d_i \right)^2 \leq 16q^2.$$

Charging every active label degree $4q$ separately would be an unnecessary extra factor of q ; the displayed summation avoids that loss.

3. Control both sides of the centered form.

Equation (4.14) and the last estimate give $-kI \preceq \mathcal{R} \preceq B_q I$. Since $B_q \geq q \geq k$, its operator norm is at most B_q . Compression by the cutoff preserves the form inequalities. Finally sum over orthogonal exact-support sectors. On the vacuum sector $S = \emptyset$, the operator vanishes. $\square$

4.4 B4. Propagate the vacuum rather than the worst state

Let $K = \min(q, m)$ and define the nonnegative symmetric tridiagonal matrix G_q on indices $0, \dots, K$ by

$$\begin{aligned} (G_q)_{00} &= 0, \quad (G_q)_{kk} = B_q(r; \lambda)/m \quad (k \geq 1), \\ (G_q)_{k+1,k} &= (G_q)_{k,k+1} = \frac{\sqrt{(k+1)(m-k)v_r}}{m} \quad (0 \leq k < K). \end{aligned} \quad (4.15)$$

If $n(f) = (\|P_{\mathcal{K}_k} f\|)_{k=0}^K$ is the vector of sector norms, Lemmas 4.2 and 4.4 give

$$n(M_q f) \leq G_q n(f) \quad (4.16)$$

componentwise. Since G_q is entrywise nonnegative, iteration from $n(e_a \otimes \Omega) = e_0$ yields

$$\boxed{\mathbb{E} \operatorname{tr} Z^{2q} \leq r \|G_q^q e_0\|_2^2 = r (G_q^{2q})_{00}.} \quad (4.17)$$

This is the framework’s nonnegative vacuum-majorant rule, with the one-step operator estimates now supplying its coefficients [1]. All occurrences of e_0 in scalar majorants refer to the active-count vacuum; it is distinct from the external basis vectors e_a .

4.4.1 A closed-form consequence

Let

$$c_q = ((2q - 1)!!)^{1/(2q)}, \quad \eta_q = (r/\delta)^{1/(2q)}. \quad (4.18)$$

Theorem 4.5 (Untruncated closed-form moment bound). *For every integer $q \geq 1$ and every $\lambda > 0$,*

$$\boxed{(\mathbb{E} \operatorname{tr} Z^{2q})^{1/(2q)} \leq r^{1/(2q)} \left[c_q \sqrt{\frac{v_r}{m}} + \frac{B_q(r; \lambda)}{m} \right].} \quad (4.19)$$

Consequently

$$\mathbb{P} \left\{ \|Z\| > \eta_q \left[c_q \sqrt{\frac{v_r}{m}} + \frac{B_q(r; \lambda)}{m} \right] \right\} \leq \delta. \quad (4.20)$$

Proof. 1. Dominate the sector matrix entrywise.

Let J_q be the $(q+1) \times (q+1)$ Hermite Jacobi matrix with zero diagonal and $(J_q)_{k+1,k} = (J_q)_{k,k+1} = \sqrt{k+1}$. After padding G_q by zeros if $m < q$, it is entrywise at most

$$bI + aJ_q, \quad a = \sqrt{v_r/m}, \quad b = B_q(r; \lambda)/m,$$

because $(k+1)(m-k) \leq (k+1)m$ and $B_q \geq 0$ on the diagonal. The entries are nonnegative, so matrix powers preserve this comparison, and in particular $\|G_q^q e_0\| \leq \|(bI + aJ_q)^q e_0\|$.

2. Identify the comparison vacuum moment as a scalar Gaussian moment.

By (4.2), J_q is multiplication by a scalar standard Gaussian ξ compressed to Hermite degrees $0, \dots, q$. Each application of $b + a\xi$ raises the degree by at most one, so q applications starting from $h_0 = 1$ never leave degrees $0, \dots, q$, and the compression changes nothing. Thus

$$\|(bI + aJ_q)^q e_0\|^2 = \mathbb{E}(b + a\xi)^{2q}.$$

3. Separate the two rates and apply Markov.

Minkowski in scalar L_{2q} gives

$$\|G_q^q e_0\|^{1/q} \leq b + a \|\xi\|_{L_{2q}} = b + ac_q.$$

Combine this with (4.17) to obtain (4.19). Finally, $\mathbb{P}\{\|Z\| > x\} \leq x^{-2q} \mathbb{E} \operatorname{tr} Z^{2q}$ at the displayed threshold proves the tail bound (4.20), since $\eta_q^{2q} = r/\delta$. $\square$

The scalar Gaussian ξ describes only the comparison Jacobi operator. It does not replace any original tensor factor or any repeated sample in the original law.

Corollary 4.6 (The common sample order follows from Chain B). *The framework proof implies (1.2). An explicit sufficient row count for any chosen $q \geq 1, \lambda > 0$ is*

$$m \geq \left\lceil \left(\frac{a_q + \sqrt{a_q^2 + 4\varepsilon b_q}}{2\varepsilon} \right)^2 \right\rceil, \quad a_q = \eta_q c_q \sqrt{v_r}, \quad b_q = \eta_q B_q(r; \lambda). \quad (4.21)$$

Proof. 1. The exact threshold.

Equation (4.21) solves $a_q/\sqrt{m} + b_q/m \leq \varepsilon$ exactly as a quadratic inequality in $\sqrt{m}$.

2. The sample-size order.

Take $q = \lceil \log(r/\delta) \rceil \geq 1$ and $\lambda = 2$. Then $\eta_q \leq \sqrt{e}$, $c_q \leq \sqrt{2q}$, and $q \leq 2t$ for $t = \log(4r/\delta) > 1$. The allowance is therefore bounded by an absolute constant times

$$\sqrt{rq/m} + rq/m + q^2/m.$$

Making each term at most a fixed fraction of ε requires the sample order $\varepsilon^{-2}rt + \varepsilon^{-1}rt + \varepsilon^{-1}t^2$. Since $\varepsilon < 1$, the middle term is absorbed by the first, which gives (1.2). $\square$

4.5 A stronger certificate: use the budget at each stage

The fixed cutoff at degree $4q$ is exact, but its largest repeated-visit cost need not be charged at every earlier step. A prefix of length $j - 1$ only needs the next cutoff at degree $4j$ and at most j active samples.

Fix the terminal q and $K = \min(q, m)$. Define a symmetric nonnegative matrix $G^{[j]}$ on $0, \dots, K$ for $1 \leq j \leq q$ by

$$\begin{aligned} (G^{[j]})_{00} &= 0, & (G^{[j]})_{kk} &= B_j(r; \lambda)/m \quad (k \geq 1), \\ (G^{[j]})_{k+1,k} &= (G^{[j]})_{k,k+1} = \frac{\sqrt{(k+1)(m-k)}v_r}{m}. \end{aligned} \quad (4.22)$$

Entries beyond currently reachable sectors are harmless padding. Propagate in the stated order:

$$w^{(0)} = e_0, \quad w^{(j)} = G^{[j]}w^{(j-1)}. \quad (4.23)$$

Theorem 4.7 (Stage-dependent vacuum certificate). *For the original independent Gaussian law,*

$$\boxed{\mathbb{E} \operatorname{tr} Z^{2q} \leq r \|G^{[q]}G^{[q-1]} \dots G^{[1]}e_0\|_2^2.} \quad (4.24)$$

In particular, the computable inequality

$$r \|G^{[q]} \dots G^{[1]}e_0\|_2^2 \leq \delta \varepsilon^{2q} \quad (4.25)$$

is sufficient for $\mathbb{P}\{\|Z\| > \varepsilon\} \leq \delta$. At fixed q, λ, r, m , this certificate is no worse than the stationary certificate (4.17).

Proof. 1. One exact stage.

For one external seed, put $f_j = \mathcal{M}_Z^j(e_a \otimes \Omega)$. By (4.3), f_{j-1} belongs to the cutoff with degree $4(j-1)$ and active count at most $j-1$, while f_j lies in the next cutoff. Let $M_j = \Pi_j \mathcal{M}_Z \Pi_j$ use that next cutoff. Then $f_j = M_j f_{j-1}$ exactly.

2. Propagate sector norms in the stated order.

Apply Lemmas 4.2 and 4.4 with parameter j :

$$n(f_j) \leq G^{[j]} n(f_{j-1}).$$

Induction gives $n(f_q) \leq w^{(q)}$ componentwise. Sum squared norms over all r external seeds and use (4.1) to obtain (4.24). Markov then yields (4.25).

3. Compare with the stationary certificate.

Finally $B_j(r; \lambda) \leq B_q(r; \lambda)$ implies $G^{[j]} \leq G_q$ entrywise. Positivity and induction give $w^{(q)} \leq G_q^q e_0$ componentwise. $\square$

Remark 4.8 (Order and energy are essential). The matrices $G^{[j]}$ generally do not commute. The terminal quantity is the squared norm of an *ordered product*; it is not a product of top eigenvalues, a power of an average, or an unordered product of return costs.

To express the argument as an energy, define for a nonnegative sector vector z

$$\mathcal{E}_j(z) = \|G^{[q]} \dots G^{[j+1]} z\|_2^2, \quad \mathcal{E}_q(z) = \|z\|^2.$$

The componentwise step inequality and nonnegativity give $\mathcal{E}_j(n(f_j)) \leq \mathcal{E}_{j-1}(n(f_{j-1}))$. This is the framework's weighted-energy viewpoint, with a specified finite certificate rather than a formal re-expression of the unknown exact moment.

4.6 An exact limited-independence consequence

Corollary 4.9 (Moment matching of sample blocks). *Fix q . Suppose each block (g_i, h_i) has the correct two-factor Gaussian marginal law, and any set of at most $2q$ distinct blocks has the same joint law as under full independence. Then $\mathbb{E} \operatorname{tr} Z^{2q}$ is unchanged. All three framework certificates at this q therefore remain valid.*

Proof. Expand $\operatorname{tr}(\sum_i X_i)^{2q}$. Each word uses at most $2q$ distinct block labels. Its expectation is fixed by their joint law, even if a label is repeated many times within the word. Sum the finitely many equal expectations and divide by m^{2q} . The operator proof is carried out on the independent reference law; only its resulting moment is transferred. $\square$

This does not provide a finite-seed Gaussian generator or transfer the lower bound. No concentration or cutoff argument on the dependent law is asserted.

5 A common matching lower bound

The two upper proofs concern the same distribution, so they share one lower-bound argument. A single large sample suffices; no spectral independence of matrix entries is required.

Theorem 5.1 (Common-factor obstruction). *Take $n_1 = r$, $n_2 = 2$, and $Ue_a = e_a \otimes e_1$ for $1 \leq a \leq r$. For all sufficiently large r , uniformly over integers $1 \leq m \leq \frac{1}{4}r \log r$,*

$$\mathbb{P}\{\|Z\| > 1/2\} \longrightarrow 1 \quad (r \longrightarrow \infty).$$

Proof. For this subspace,

$$y_i = \eta_i \xi_i, \quad \eta_i \sim N(0, 1), \quad \xi_i \sim N(0, I_r),$$

with all scalar and vector variables independent.

1. Below r samples, rank already forces failure.

If $m < r$, the sample covariance $m^{-1} \sum_i y_i y_i^\top$ has rank at most m , so it has a zero eigenvalue. Therefore $\|Z\| \geq 1$ deterministically.

2. Select the large scalar without biasing its vector factor.

Suppose $m \geq r$ and let J be the smallest index maximizing η_i^2 . The index depends only on the scalars. Conditional on all of them, ξ_J still has law $N(0, I_r)$; thus it is independent of those scalars. Positivity of the remaining covariance summands gives

$$\lambda_{\max} \left(\frac{1}{m} \sum_i y_i y_i^\top \right) \geq \frac{\eta_J^2 \|\xi_J\|^2}{m}. \quad (5.1)$$

3. Lower-bound the selected scalar and the vector length.

For $m \geq 3$, integrate the Gaussian density on $[a, a + 1/a]$ and its reflection, where $a = \sqrt{\log m} \geq 1$. Since $(a + 1/a)^2 \leq a^2 + 3$, this gives an absolute $c_1 > 0$ such that

$$p_m := \mathbb{P}\{\eta^2 \geq \log m\} \geq c_1 \frac{m^{-1/2}}{\sqrt{\log m}}.$$

Independence implies

$$\mathbb{P}\{\eta_J^2 < \log m\} = (1 - p_m)^m \leq \exp \left(-c_1 \frac{\sqrt{m}}{\sqrt{\log m}} \right).$$

Also $\mathbb{E}e^{-\|\xi\|^2/2} = 2^{-r/2}$, so exponential Markov gives

$$\mathbb{P}\{\|\xi_J\|^2 < r/2\} \leq e^{r/4} 2^{-r/2} \leq e^{-r/16}.$$

Consequently, with probability at least

$$1 - \exp \left(-c_1 \frac{\sqrt{m}}{\sqrt{\log m}} \right) - e^{-r/16}, \quad (5.2)$$

the right side of (5.1) is at least $r \log m / (2m)$.

4. Read off the necessary sample order.

If $r \leq m \leq \frac{1}{4}r \log r$, then $\log m \geq \log r$, and hence $r \log m / (2m) \geq 2$. On the event above, $\|Z\| \geq 1 > 1/2$. The probability in (5.2) tends to one uniformly throughout this range, since $m \geq r \rightarrow \infty$. Together with the rank argument this proves the claim. $\square$

At $\varepsilon = 1/2$, $\delta = 0.01$, one has $t = \log(400r) = O(\log(er))$ and $t^2 = O(r \log(er))$. Either upper proof and Theorem 5.1 therefore establish the $\Theta(r \log(er))$ assertion of Theorem 1.1. The obstruction is worst-case: a particular subspace may need fewer rows.

6 Comparing end results and explicit constants

6.1 The four certificates being compared

To avoid an unfair comparison between an unspecified constant and an explicit one, we compare only the following completely specified allowances:

$$U_A(m, r, \delta) = \sqrt{\frac{18rt}{m}} + \frac{144(r+s)t}{m} + 3e^{-s}, \quad t = \log(4r/\delta), \quad s = \lceil \log(4m/\delta) \rceil; \quad (6.1)$$

$$U_{B,\text{cl}}(m, r, q, \lambda, \delta) = \eta_q \left[c_q \sqrt{\frac{v_r}{m}} + \frac{B_q(r; \lambda)}{m} \right]; \quad (6.2)$$

$$U_{B,\text{st}}(m, r, q, \lambda, \delta) = \left[\frac{r}{\delta} \|G_q^q e_0\|_2^2 \right]^{1/(2q)}; \quad (6.3)$$

$$U_{B,\text{dyn}}(m, r, q, \lambda, \delta) = \left[\frac{r}{\delta} \|G^{[q]} \dots G^{[1]} e_0\|_2^2 \right]^{1/(2q)}. \quad (6.4)$$

Each bound being at most ε certifies the desired probability. At the *same* m, r, q, λ, δ ,

$$U_{B,\text{dyn}} \leq U_{B,\text{st}} \leq U_{B,\text{cl}}. \quad (6.5)$$

The first inequality preserves the smaller degree budgets of earlier steps; the second is the shifted-Gaussian and Minkowski relaxation of the actual vacuum. There is no comparable universal ordering between U_A and the B -certificates.

Both chains yield (1.2). Chain B directly gives all-order, untruncated trace moments without a $\log m$ term. Chain A's explicit tail bound contains $\log m$ through s , but Proposition 3.6 absorbs it into the same sufficient sample order. This is a difference in proof and moment output, not an improvement in the asymptotic rate.

6.2 Leading constants: what agrees and what does not

Let $h = \log(r/\delta)$ and $q = \lceil h \rceil$, with $h \rightarrow \infty$. Stirling's formula applied to $(2q-1)!! = (2q)!/(2^q q!)$ gives

$$c_q \sim \sqrt{2q/e}, \quad \eta_q \sim \sqrt{e}.$$

The fresh-sample term of $U_{B,\text{cl}}$ is therefore

$$\eta_q c_q \sqrt{v_r/m} \sim \sqrt{2v_r h/m}. \quad (6.6)$$

When also $r \rightarrow \infty$, $v_r = 9r - 1 \sim 9r$, so this agrees with Chain A's leading $\sqrt{18rh/m}$ term, since $t = h + \log 4$.

For comparison, Schur's row-sum bound on the Gaussian Jacobi matrix gives $\|J_q\| \leq 2\sqrt{q}$. Replacing the vacuum moment by this uniform norm would charge $2\sqrt{qv_r/m}$ before Markov rather than $c_q \sqrt{v_r/m}$. The ratio of these two explicit fresh-sample allowances tends to

$$\frac{2\sqrt{q}}{c_q} \longrightarrow \sqrt{2e}. \quad (6.7)$$

This is an identifiable loss from the full-norm relaxation, not a claim that an operator's actual norm equals its Schur bound.

The closed-form repeated-visit coefficient is less favorable. With $\lambda = 2$, $q \sim h$, and $r/h \rightarrow \infty$, the coefficient of rh/m in $U_{B,\text{cl}}$ tends to

$$99e^{3/2} \approx 443.69.$$

In U_A it is 144 when additionally $s/r \rightarrow 0$. Thus the simple closed-form coefficient is about 3.08 times as large. That does *not* imply a universal 3.08 ratio between sample counts: the variance term and all other terms must also be included.

The repeated-visit proof visibly incurs a factor 3 for three-degree overlap, a factor $e^{2/\lambda}$ from Hölder/hypercontractivity, and the relaxations $\sum_i d_i \leq 4q$, $\sum_i d_i^2 \leq 16q^2$. The ordered certificate retains what can still be recovered after those local estimates: the zero vacuum diagonal, the finite-sample activation factors, and the smaller earlier degree budgets.

6.3 Numerical sufficient row counts, with all parameters specified

The following table uses $\varepsilon = 1/2$ and $\delta = 0.01$. Every entry is a selected sufficient integer m , rounded upward with a margin and substituted back into its certificate. The sample counts are not numerical estimates of the actual random sketch’s performance, and no minimum over all possible proof parameters is claimed.

r	Chain A	Chain B		
	Concentration	Closed form	Stationary vacuum	Stage-dependent
10	91 000	194 000	65 000	60 000
100	580 000	1 280 000	630 000	472 000
1 000	6 200 000	12 610 000	7 260 000	5 150 000
10 000	72 000 000	145 300 000	90 100 000	61 000 000

Table 1: Conservative sufficient theorem certificates. Columns use (6.1), (6.2), (6.3), and (6.4), respectively. They are not optimal or empirical row counts.

r	Closed form (q, λ)	Stationary (q, λ)	Stage-dependent (q, λ)
10	(3, 1)	(2, 1)	(3, 1)
100	(4, 2)	(4, 2)	(5, 2)
1 000	(7, 2)	(5, 2)	(7, 2)
10 000	(8, 2)	(7, 2)	(8, 2)

Table 2: Proof parameters for Table 1. Chain A has no q or λ parameter; t and s are set by (6.1).

Appendix C records conservative upward-rounded distortion allowances at all these sample counts. The accompanying verification script recomputes them with 80-digit arithmetic and 70-digit outward-rounded intervals. At $r = 1\,000$, the stage-dependent certificate uses about 17% fewer rows than the specified concentration baseline; at $r = 10\,000$, about 15% fewer. These comparisons concern only the displayed certificates. The constant 72 in Chain A is conservative, and both chains admit further optimization.

End result: the framework route matches the sample-size order and gives an original-law trace-moment theorem. Its coarse closed form has larger repeated-visit constants. Keeping the actual, stage-dependent vacuum propagation improves the stated numerical certificates, including relative to the chosen explicit concentration baseline. This is not a universal superiority claim.

7 Scope, provenance, and verification status

The factors throughout are independent real standard Gaussian vectors, and the subspace is deterministic. No step establishes the same improvement for arbitrary isotropic sub-Gaussian factor vectors, higher tensor orders, or subspaces chosen after observing the sketch. The lower bound is worst-case, not a classification of every subspace by Schmidt geometry. Optimal joint ε, δ dependence is not claimed.

Chain A reflects the coefficient-retention viewpoint but is a standalone concentration argument. Chain B uses the framework’s exact multiplication, graded-domain reduction, coefficient-sensitive block estimates, and vacuum/energy propagation. It does not need the positive-coefficient cone, the graph connected-creation theorem, general encoding/descent tests, signed conditional cores, or covariance-spectrum renewal. In particular, its Gaussian hypercontractivity bound applies to arbitrary polynomial coefficients.

The starting material was the two proof notes developed in this conversation: *Sharp fixed-confidence order-two Gaussian Khatri–Rao embeddings* and *Gaussian Khatri–Rao embeddings through the graded multiplication pipeline*, both dated 16 September 2026. This document consolidates their application arguments, normalizations, lower bound, and explicit constants into one text. The organizing framework is the supplied v11 manuscript [1]. The Gaussian inequalities and matrix-analysis inputs are classical and are credited separately [2, 3, 4, 5].

The accompanying checks verify exact finite scalar inequalities, selected high-precision numerical certificates, and normalization benchmarks. They do not constitute an independent proof audit, a Lean certificate, a test of every possible parameter value, or a claim of publication priority. The general statements are justified by the analytic proofs, not by the numerical tests.

A Gaussian inputs and Hermite normalization

This appendix records short standard semigroup arguments for the Gaussian inequalities used in Section 2. They are included to fix constants, not asserted as new results; see Gross [2, 3].

A.1 Ornstein–Uhlenbeck semigroup and logarithmic Sobolev inequality

For a smooth function F on $\mathbb{R}^N$, let

$$(\mathbf{P}_t F)(x) = \mathbb{E} F(e^{-t}x + \sqrt{1 - e^{-2t}}G), \quad G \sim N(0, I_N).$$

Standard Gaussian measure is invariant under $\mathbf{P}_t$; this follows directly because a sum of independent centered Gaussian vectors with squared coefficients summing to one is standard Gaussian. Differentiation of the kernel gives the generator $\mathcal{L} = \Delta - x \cdot \nabla$, and Gaussian integration by parts gives

$$\mathbb{E} f \mathcal{L} g = -\mathbb{E} \langle \nabla f, \nabla g \rangle, \quad \nabla \mathbf{P}_t F = e^{-t} \mathbf{P}_t \nabla F.$$

1. Differentiate the entropy along the semigroup.

Assume first that F is smooth, bounded, and bounded below by a positive constant. Set $u_t = \mathbf{P}_t F$. Since $\mathbb{E} u_t = \mathbb{E} F$,

$$\frac{d}{dt} \text{Ent}(u_t) = \mathbb{E}(\mathcal{L} u_t) \log u_t = -\mathbb{E} \frac{\|\nabla u_t\|^2}{u_t}.$$

As $t \rightarrow \infty$, $u_t \rightarrow \mathbb{E} F$ and its entropy tends to zero, so integrating this derivative recovers $\text{Ent}(F)$.

2. Bound the dissipation pointwise.

The gradient commutation identity and weighted Cauchy–Schwarz give pointwise

$$\frac{\|\nabla \mathbf{P}_t F\|^2}{\mathbf{P}_t F} \leq e^{-2t} \mathbf{P}_t \left(\frac{\|\nabla F\|^2}{F} \right).$$

3. Integrate in space, then in time.

Integrate in Gaussian measure, use invariance, and then integrate over t :

$$\text{Ent}(F) \leq \int_0^\infty e^{-2t} dt \mathbb{E} \frac{\|\nabla F\|^2}{F} = \frac{1}{2} \mathbb{E} \frac{\|\nabla F\|^2}{F}.$$

Taking $F = f^2$ gives

$$\text{Ent}(f^2) \leq 2 \mathbb{E} \|\nabla f\|^2,$$

which is (2.1). Positive regularization, truncation, and smooth approximation extend the inequality to the functions with finite energy used in the main proof.

A.2 Hypercontractivity and homogeneous chaos

1. A monotone norm along the semigroup.

Let $f > 0$ be smooth and bounded, $u_t = \mathbf{P}_t f$, and $p(t) = 1 + (p_0 - 1)e^{2t}$ with $p_0 > 1$. Differentiation yields

$$\frac{d}{dt} \log \|u_t\|_{L_{p(t)}} = \frac{p'(t)}{p(t)^2} \frac{\text{Ent}(u_t^{p(t)})}{\mathbb{E} u_t^{p(t)}} - (p(t) - 1) \frac{\mathbb{E} u_t^{p(t)-2} \|\nabla u_t\|^2}{\mathbb{E} u_t^{p(t)}}.$$

Apply (2.1) to $u_t^{p(t)/2}$:

$$\text{Ent}(u_t^{p(t)}) \leq \frac{p(t)^2}{2} \mathbb{E} u_t^{p(t)-2} \|\nabla u_t\|^2.$$

Since $p'(t) = 2(p(t) - 1)$, the derivative of the logarithmic norm is nonpositive. Thus $\|P_t f\|_{L_{p(t)}} \leq \|f\|_{L_{p_0}}$. Positivity of the semigroup and $|P_t f| \leq P_t |f|$ extend this to signed functions, then approximation extends it to the relevant L_p spaces.

2. Specialize to homogeneous chaos.

The Hermite generating function

$$e^{zx - z^2/2} = \sum_{\ell=0}^{\infty} \text{He}_{\ell}(x) \frac{z^{\ell}}{\ell!}$$

shows that P_t acts on homogeneous total degree d by e^{-dt} . Take $p_0 = 2$ and $u = 1 + e^{2t}$. For a scalar homogeneous Hermite polynomial F_d ,

$$e^{-dt} \|F_d\|_{L_u} \leq \|F_d\|_{L_2}, \quad \text{hence} \quad \|F_d\|_{L_u} \leq (u - 1)^{d/2} \|F_d\|_{L_2}.$$

3. Allow Hilbert-space coefficients.

For a Hilbert-valued polynomial with orthonormal coordinates $F_{d,a}$, Minkowski in $L_{u/2}$ gives

$$\|F_d\|_{L_u}^2 = \left\| \sum_a |F_{d,a}|^2 \right\|_{L_{u/2}} \leq \sum_a \|F_{d,a}\|_{L_u}^2 \leq (u - 1)^d \sum_a \|F_{d,a}\|_{L_2}^2.$$

Finite-dimensional projections and monotone convergence justify arbitrary Hilbert coefficient spaces. This proves (2.2) with the stated constant.

A.3 The scalar Jacobi comparison is exact through the needed moment

For $h_{\ell} = \text{He}_{\ell} / \sqrt{\ell!}$, the generating function also yields (4.2). The first $q + 1$ Hermite functions therefore represent compressed scalar multiplication by $\xi \sim N(0, 1)$ with Jacobi matrix J_q . An induction on the prefix length gives

$$(bI + aJ_q)^j e_0 \longleftrightarrow (b + a\xi)^j, \quad 0 \leq j \leq q.$$

The Gaussian moment recurrence, obtained by integration by parts, is $\mathbb{E}\xi^{2q} = (2q - 1)\mathbb{E}\xi^{2q-2}$. It follows that

$$\|J_q^q e_0\|^2 = (2q - 1)!!, \quad \|(bI + aJ_q)^q e_0\|^2 = \mathbb{E}(b + a\xi)^{2q}.$$

These are vacuum identities, not traces of the entire Jacobi matrix.

B An all-parameter verification of the constant 72

We prove (3.3) without optimizing the constant. Set

$$a_p = 8p - 11, \quad b_p = (4p - 6)^2, \quad F_p(r) = \frac{(a_p r + b_p)^{p-1}}{r(r + p)^{p-2}}.$$

For integer $p \geq 3$, the sign of the logarithmic derivative in r is the sign of

$$[-8p^3 + 37p^2 - 62p + 36]r - b_p p.$$

Indeed, multiply $(p - 1)a_p/(a_p r + b_p) - 1/r - (p - 2)/(r + p)$ by its positive common denominator and collect terms. If $z = p - 3 \geq 0$, the bracket becomes

$$-8z^3 - 35z^2 - 56z - 33 < 0.$$

Thus $F_p(r)$ decreases for $r \geq 1$, so it is enough to prove the inequality at $r = 1$. Since $a_p + b_p = (4p - 5)^2$, the ratio of its left to right sides is

$$\rho_p = \frac{2(4p - 5)^{2p-2}}{3p! 72^{p-2} (p + 1)^{p-2}}. \quad (\text{B.1})$$

Finite orders $3 \leq p \leq 10$.

The following rational upper bounds are verified by integer cross-multiplication in (B.1); each is less than one.

p	3	4	5	6	7	8	9	10
Upper bound on ρ_p	927/1000	380/1000	177/1000	88/1000	46/1000	25/1000	14/1000	8/1000

For example $\rho_3 = 2401/2592 < 927/1000$. All eight exact fractions are also recorded by the companion verification script.

All remaining orders $p \geq 11$.

The elementary inequality $p! \geq (p/e)^p$ follows from $\sum_{k=1}^p \log k \geq \int_1^p \log x \, dx$. Using $4p - 5 \leq 4p$ and $p + 1 \geq p$ in (B.1) gives

$$\rho_p \leq \frac{32e^2}{3} \left(\frac{2e}{9} \right)^{p-2}.$$

Since $e < 11/4$ and $11/18 < 1$, for every $p \geq 11$,

$$\rho_p < \frac{242}{3} \left(\frac{11}{18} \right)^9 = \frac{285311670611}{297538935552} < 1.$$

This proves (3.3) for every real $r \geq 1$ and integer $p \geq 3$. It is an all-parameter analytic certificate with only eight finite integer checks, not a numerical sweep over dimensions or high moment orders.

C Numerical certificates and reproducibility

C.1 A direct recurrence, without matrix-power software

For $K = \min(q, m)$, precompute

$$a_k = \frac{\sqrt{(k+1)(m-k)(9r-1)}}{m}, \quad 0 \leq k < K.$$

Set $w_0^{(0)} = 1$ and $w_k^{(0)} = 0$ for $k \geq 1$. For $j = 1, \dots, q$, use $b_j = B_j(r; \lambda)/m$ for the stage-dependent certificate, or $b_j = B_q(r; \lambda)/m$ for the stationary one, and update

$$\begin{aligned} w_0^{(j)} &= a_0 w_1^{(j-1)}, \\ w_k^{(j)} &= a_{k-1} w_{k-1}^{(j-1)} + b_j w_k^{(j-1)} + a_k w_{k+1}^{(j-1)}, \quad 1 \leq k \leq K. \end{aligned} \tag{C.1}$$

Terms with indices outside $0, \dots, K$ are zero; in particular the final term at $k = K$ is absent. This recurrence takes $O(q \min(q, m))$ arithmetic operations and $O(\min(q, m))$ stored numbers. The final failure certificate is

$$F = \frac{r}{\varepsilon^{2q}} \sum_{k=0}^K (w_k^{(q)})^2.$$

The condition $F \leq \delta$ is exactly (4.25) or its stationary analogue. Each operation is monotone on nonnegative inputs, so upward-rounded interval evaluation gives an upper bound for the entire recurrence.

C.2 Actual allowance bounds at the displayed row counts

Table 3 lists conservative six-decimal upper bounds for the allowances at the sample counts and proof parameters in Tables 1–2. Each number is rounded upward, not to nearest, and is below $\varepsilon = 0.5$.

r	Chain A	Closed form	Stationary vacuum	Stage-dependent
10	0.495576	0.498777	0.499382	0.494373
100	0.497052	0.498401	0.499537	0.498858
1 000	0.499705	0.499817	0.499886	0.499417
10 000	0.499744	0.499816	0.499835	0.499431

Table 3: Upward-rounded distortion allowances at the sufficient row counts. The certificates, not random simulations, produce these numbers.

The companion script `khatri_rao_two_proofs_checks.py` evaluates the certificates at 80 decimal digits and verifies the inequalities with 70-digit outward-rounded intervals. For Chain A it also encloses $\log(4m/\delta)$ between two consecutive integers before fixing its ceiling. The associated JSON stores the parameters, exact finite fractions, computed allowances, and check totals. Running the script requires Python 3.10 or later and `mpmath`; the recurrence above suffices for an independent implementation.

C.3 Exact scalar and normalization checks

For the scalar common-factor model, let $X = g^2 h^2 - 1$. With $(-1)!! = 1$,

$$\mu_k := \mathbb{E}X^k = \sum_{\ell=0}^k \binom{k}{\ell} (-1)^{k-\ell} [(2\ell-1)!!]^2. \quad (\text{C.2})$$

If $M_k^{(j)} = \mathbb{E}(X_1 + \cdots + X_j)^k$, independence yields the exact recurrence

$$M_k^{(j+1)} = \sum_{\ell=0}^k \binom{k}{\ell} M_{k-\ell}^{(j)} \mu_\ell, \quad M_0^{(0)} = 1, \quad M_k^{(0)} = 0 \quad (k > 0).$$

Dividing $M_{2q}^{(m)}$ by m^{2q} gives an exact rational scalar trace moment. The script checks it against the stage-dependent majorant for $m \in \{1, 2, 5\}$ and $1 \leq q \leq 6$, giving 18 checks. At $q = 1$, both sides equal $8/m$ exactly because $\mathbb{E}X^2 = 9 - 1 = 8$. It separately checks $\|J_q^q e_0\|^2 = (2q-1)!!$ for $1 \leq q \leq 8$.

In total the reported run verifies 16 interval sample certificates, 18 exact scalar-moment comparisons, eight Jacobi normalizations, eight finite integer inequalities for the moment scale, and one all-orders rational envelope. The scalar tests are useful indexing and normalization checks; they are not proofs for arbitrary subspaces. That generality comes from the analytic coefficient and operator estimates in the main text.

References

- [1] D. Heidary, *Graded Multiplication Operators for Random-Matrix Moments*, Version 11, exposition and attribution revision, 16 September 2026. Supplied manuscript `v11_attribution_revised.tex`. Relevant principles: exact finite moment compression; nonnegative vacuum majorants; weighted energy transport. Their application here is proved in full.
- [2] L. Gross, *Logarithmic Sobolev inequalities*, American Journal of Mathematics **97** (1975), 1061–1083. DOI: [10.2307/2373688](https://doi.org/10.2307/2373688).
- [3] L. Gross, *Logarithmic Sobolev inequalities and contractivity properties of semigroups*, in G. Dell’Antonio and U. Mosco (eds.), *Dirichlet Forms*, Lecture Notes in Mathematics **1563**, Springer, 1993, 54–88. DOI: [10.1007/BFb0074091](https://doi.org/10.1007/BFb0074091).

- [4] E. H. Lieb, *Convex trace functions and the Wigner–Yanase–Dyson conjecture*, *Advances in Mathematics* **11** (1973), 267–288, Theorem 6. The trace-concavity consequence used here is stated in (3.8); see also Tropp, Theorem 3.2.
- [5] J. A. Tropp, *User-friendly tail bounds for sums of random matrices*, *Foundations of Computational Mathematics* **12** (2012), 389–434. [arXiv:1004.4389v7](#), Theorem 3.2, Corollary 3.3, and Lemma 3.4.
- [6] L. Beretta and C. Musco, *A Tight Analysis of Khatri–Rao Oblivious Subspace Embeddings*, 2026. [arXiv:2608.28094v2](#), Theorem 1.3. The comparison in this note concerns only its order-two Gaussian specialization.